**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 181 · Recursion: base cases and returns

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. A C# function is declared int Double(int n). What does return n * 2; do when n is 4?

Q2. Starting total = 1, repeat total = total * i for i = 1, 2, 3. What is total after each iteration?

Q3. Values A, B and C are pushed onto a stack in that order. Which leaves first when popping?

Q4. A function AddTwo(x) returns x + 2. Calculate AddTwo(AddTwo(3)) and show the inner and outer calls.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

recursion — A function calling itself to solve a smaller problem.

base case — A condition that stops further recursive calls.

recursive case — A call to the same function with a smaller or simpler input.

call stack — Memory tracking active function calls and their local state.

return value — The result passed back to a caller.

## New learning

Start with a small input — Factorial(4) calls Factorial(3); it cannot finish multiplying until the inner call returns.

Reach the stopping rule — Factorial(0) returns 1. It makes no further recursive call.

Return through the calls — Factorial(1) = 1 × 1; Factorial(2) = 2 × 1; Factorial(3) = 3 × 2; Factorial(4) = 4 × 6 = 24.

## Worked example

Trace Factorial(3). Show each call and its return value.

1. Calls — 3 → 2 → 1 → 0. The argument decreases by one each time.

2. Base case — Factorial(0) returns 1. No more calls occur.

3. Returns — Factorial(1)=1, Factorial(2)=2, Factorial(3)=6. Final answer: 6.

# We Do · Practise with guidance

Trace Factorial(4) without running the program. Identify the base case and final returned value.

Factorial(4) → ... → Factorial(0)

Predict the next step before revealing the explanation.

Predict the next call

4 calls 3; 3 calls 2; 2 calls 1; 1 calls 0.

Predict the first return

Factorial(0) returns 1.

Complete the return chain

Factorial(1)=1, Factorial(2)=2, Factorial(3)=6, Factorial(4)=24.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Complete the calls and returns for Factorial(2).

### Question 2

What happens if Factorial has no base case? Explain using the call stack.

### Question 3

Modify Factorial to reject negative input; write the C# guard and explain the reason.

# Exit Ticket · Answer before checking

If a recursive call reaches its base case, all previous calls have already returned.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. It sends the value 8 back to the caller. A return statement also ends the current invocation.

Q2. 1, 2 and 6. Each iteration multiplies the previous total by the new i.

Q3. C. A stack is last in, first out (LIFO).

Q4. The inner AddTwo(3) returns 5; then the outer AddTwo(5) returns 7.

### You Do pages · Independent questions

Q1. Calls: 2 → 1 → 0. Return from 0 is 1, then 1 returns 1 and 2 returns 2.

Q2. Calls continue without a stopping condition, adding stack frames until the runtime eventually fails (typically stack overflow).

Q3. Before the base case: if (n < 0) throw new ArgumentOutOfRangeException(nameof(n));. Negative values would otherwise keep decreasing and never reach 0.

### Exit Ticket · Model answer

False. Previous calls are still waiting; they return as the stack unwinds.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding